

Bazooka - how it was used

PAGES FROM THE ORIGINAL MANUAL

FM 23-30, Hand and Rifle Grenades; Rocket, AT, HE,
2.36-inch (War Department, 1944)

WHAT'S INSIDE

Chapter 3: the bazooka rockets and launchers M1A1 and M9
– description, ammunition, care, operation and how the team
fought with it.

51 pages from the original follow: the manual's cover, then the section about this
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FM 23-30

WAR DEPARTMENT BASIC FIELD MANUAL

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HAND AND RIFLE GRENADES

ROCKET, AT, HE, 2.36-INCH

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1942.)

WAR DEPARTMENT • 14 FEBRUARY 1944

CHAPTER 3

ROCKET, AT, 2.36-INCH AND LAUNCHER

SECTION I

GENERAL

62. BASIC DOCTRINE GOVERNING EMPLOYMENT.

a. The antitank rocket launcher, 2.36-inch, is both an offensive and defensive weapon. In both types of action it is used primarily against hostile armored vehicles which come within effective range. It is also employed against emplaced automatic weapons, buildings, and embrasures. The weapon is essentially one of opportunity. In the hands of trained personnel, it is a powerful supporting weapon at short ranges.

b. No additional personnel are allotted for the operation of the launchers and rockets. They will be operated and properly cared for by designated personnel in addition to their regularly assigned duties. They will not ordinarily be used while men are engaged in their primary duties except when necessary to repel a mechanized attack. Rockets are supplemented by the antitank grenade in the hands of rifle grenadiers.

c. To provide necessary assistance in loading and reloading, the launcher is normally operated by a team

of two men. (See fig. 45.) In emergencies it may be operated by one man. For mutual support, it is highly desirable that teams of rocketeers work in pairs (or threes) and be so located that they may obtain flanking fire against the lighter armor on the sides or rear of hostile mechanized vehicles. Employment of launchers in pairs (or threes) will facilitate the action of combating



Figure 45. Rocket team.

two or more hostile armored vehicles which attack simultaneously from different directions.

d. As many individuals should be trained in the use of the launcher and rockets as training time and ammunition will permit.

63. BASIS OF ISSUE. The basis of issue of launchers is prescribed in the Tables of Organization and Equipment of the particular arm or service.

SECTION II

CHARACTERISTICS AND DESCRIPTION OF ANTITANK ROCKET LAUNCHERS M1A1 AND M9

64. CHARACTERISTICS. a. General. The antitank rocket launcher M1A1 or M9 is a smooth-bore, breech-loading, electrically-operated, shoulder weapon of the open-tube type. It is used to launch high-explosive rockets and is fired from the standing, kneeling, sitting, or prone positions.

b. Purpose. The primary function of the launcher is to give direction to the rocket. There is no recoil since the propulsion of the rocket is accomplished by the jet action of the propellant powder in the stabilizer tube of the rocket and does not depend upon gas pressure built up inside the launcher tube. The launcher tube, therefore, need only be heavy enough to prevent denting or bending during handling and to prevent excessive heating at normal rates of fire.

65. GENERAL DATA. General data of the antitank rocket launchers M1A1 and M9 are as follows:

Length:

M1A1 54.5 inches

M9 61 inches

Weight:

M1A1 13.26 pounds

M9 15 pounds

Diameter, internal 2.36 inches

Range:

Point targets 50 to 300 yards

Area targets 300 to 650 yards

Elevation for maximum range.....40°

66. DESCRIPTION OF LAUNCHER M1A1. a. Nomenclature. The launcher M1A1 is equipped with sights, hand grip, stock, sling, firing mechanism, face guard, breech guard tail latch, and flash deflector. (See figs. 46, 47, and 48.)

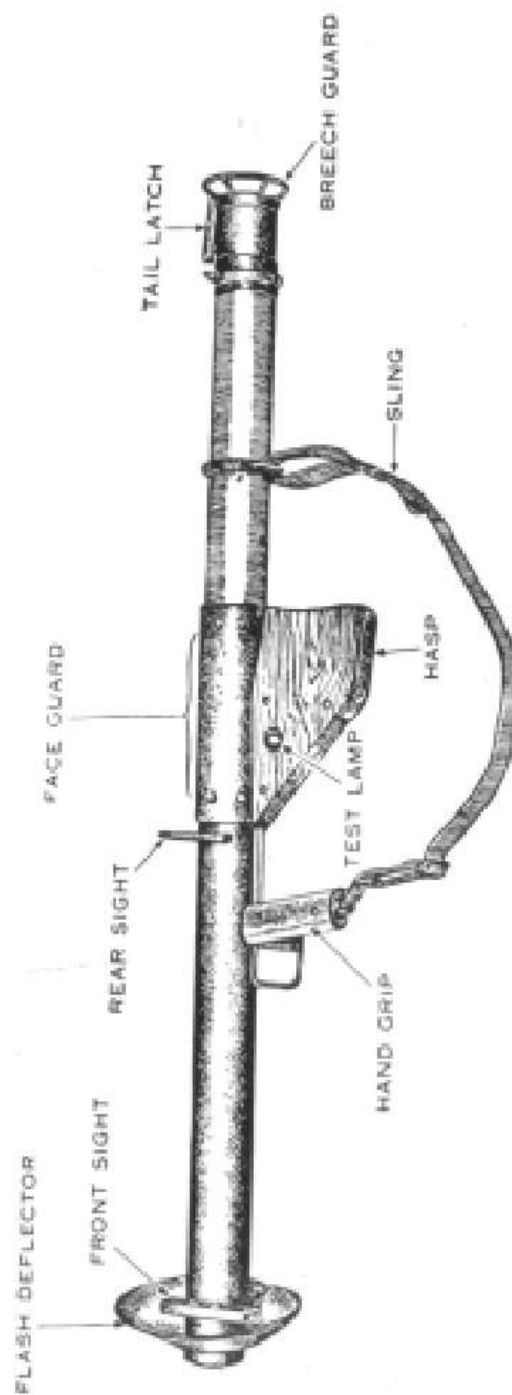
b. Component parts. (1) **SIGHTS** (fig. 47). The front and rear sights are mounted on the left side and are $20\frac{7}{8}$ inches apart. The front sight consists of three studs representing ranges of 100, 200, and 300 yards. The rear sight is a peep sight. The sights being fixed, the gunner must estimate intermediate ranges, ranges in excess of 300 yards, and the windage and lead.

(2) **HAND GRIP.** The hand grip consists of the left and right trigger grips attached to the trigger support. The trigger support accommodates the trigger guard, trigger, and the lower and upper trigger switch contacts. The trigger is pinned at its upper end to the trigger support and is free to pivot.

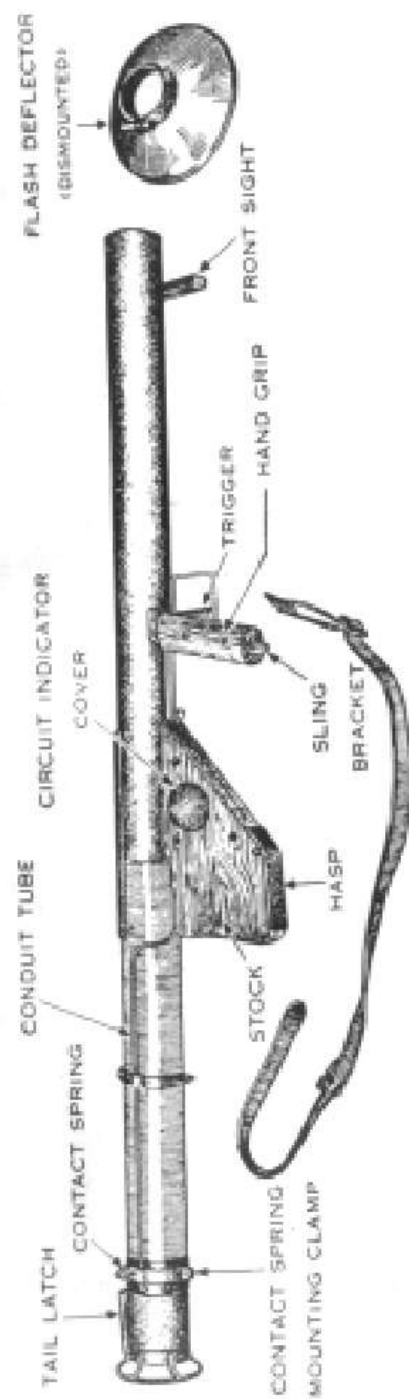
(3) **STOCK.** (a) The stock is made of wood, shaped to fit against the shoulder.

(b) In the bottom of the stock are two vertical cylindrical compartments which house four batteries. The

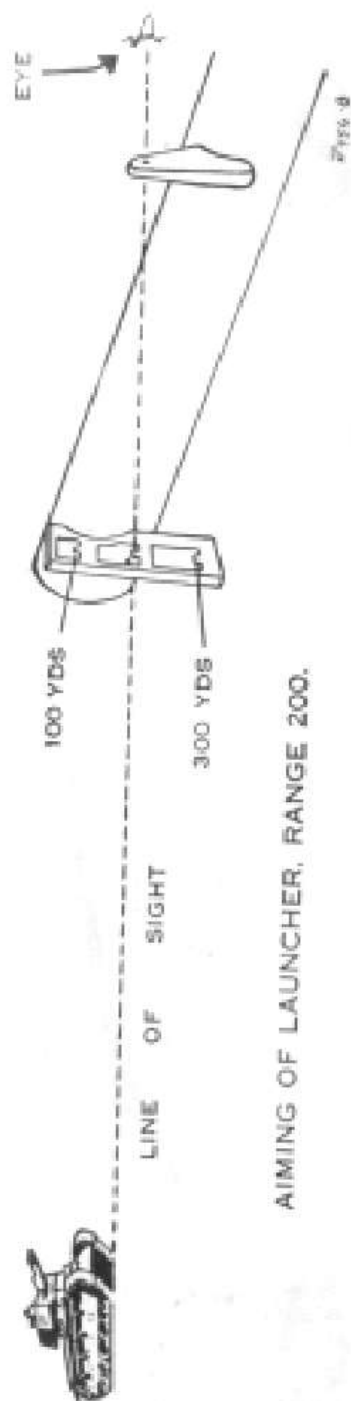
Figure 46. *Launcher M. A. r.*



① *Left side.*



© Right side.



AIMING OF LAUNCHER, RANGE 200.

Figure 47- Sights; aiming the launcher M.A.T.

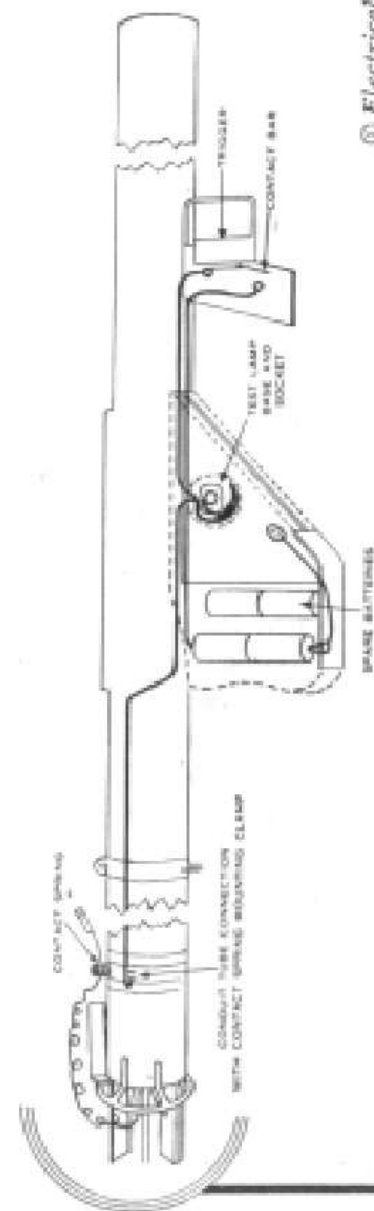
two batteries in the rear compartment supply the current to ignite the rocket; the two in the front compartment are spares. (See fig. 48①.)

(c) Contained in a recess on the left side is a small electric lamp which is used as a circuit tester. A spare lamp is carried in a circular compartment on the right side of the stock under the circuit indicator cover.

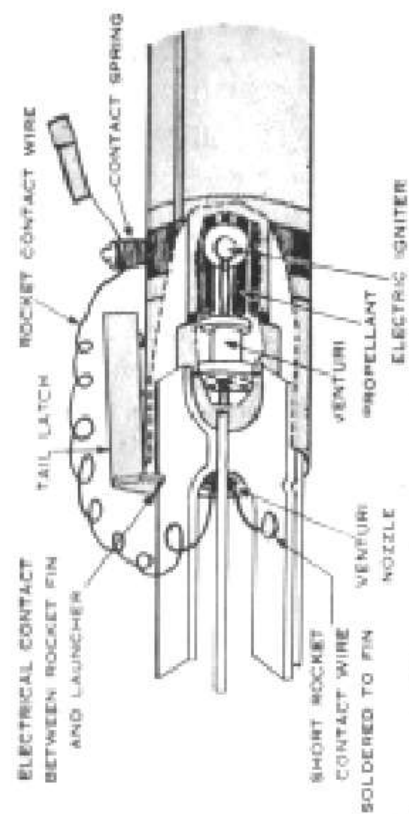
(4) SLING. The sling is fastened to the hand grip and to a bracket mounted on the launcher between the stock and the breech. It is used for carrying the launcher but is not used when firing.

(5) FIRING MECHANISM (fig. 48① and ②). (a) The bottom of the wooden stock is fitted with a hasp assembly to which is fastened a coiled battery spring. When the hasp is closed, the coiled battery spring makes contact with the negative pole of the batteries and holds them in position. The coiled battery spring is connected by a short wire to the metal stock support to complete the circuit. The positive pole of the batteries is connected by wire, running between the stock and launcher tube, forward and then down to the lower trigger switch contact. The gap between the lower trigger switch contact and the upper trigger switch contact is bridged, when the trigger is squeezed, by a copper contact bar. The wire from the upper trigger switch contact is carried toward the rear by a conduit tube mounted on the right-hand side of the launcher tube; in its passage by the stock support it is connected to the circuit-tester-lamp socket (see (b) below). It then is connected to the contact spring mounting clamp at the rear end of the launcher tube.

(b) The test lamp, set in the left side of the stock, is wired in parallel to the operating circuit to enable the firer to determine whether the battery and electrical connections are in order. The test lamp should light when the trigger is squeezed regardless of whether a rocket is in the tube.



② Electrical circuit on launcher.



③ Method of completing between rocket electrical circuit and launcher.

Figure 48. Firing mechanism wiring diagram, launcher M41.

(6) **FACE GUARD.** The face guard is a curved metal plate 10 inches long which fits closely around the launcher tube, projecting about 2 inches forward of the stock. It protects the face of the gunner when the tube becomes heated through firing. (See fig. 46③.)

(7) **BREECH GUARD.** The breech guard is a flaring metal ring mounted on the rear of the tube. It facilitates loading the rocket, prevents denting of the tube during handling, and prevents dirt and mud from entering the tube when its rear end is resting on the ground. (See fig. 46④.)

(8) **TAIL LATCH.** The tail latch is a spring-actuated latch situated on the top of the launcher tube and projecting over the breech. When a rocket is inserted in the tube, the latch engages in one or two of the notches on the tail fins of the rocket and keeps the rocket from sliding either farther into or out of the tube while the weapon is being aimed. That part of the latch which engages the notches, and the notches themselves, are unpainted in order that the electric circuit to the igniter in the stabilizer tube of the rocket will be grounded through the launcher tube. The contact edge of the tail latch and the notches are cammed to permit the automatic release of the rocket when it is fired. (See figs. 46⑤ and 48.)

(9) **FLASH DEFLECTOR.** Normally the propellant charge contained in the stabilizer tube has completely burned before the rocket leaves the launcher. Occasionally, however, especially in cold climates, the burning is somewhat retarded, creating a backflash as the rocket leaves the muzzle. The deflector prevents particles of unburned powder from blowing back onto the gunner's face and hands. The deflector is made of fine wire mesh with a rigid wire frame of a conical shape. It is fastened

to the tube by a mounting clamp with screw and nut, and is mounted flush with the muzzle end of the launcher. (See fig. 46①.)

67. DESCRIPTION OF LAUNCHER M9. a. General.

The launcher M9 is a later model than the launcher M1A1 and is standard. It has a two-piece barrel; for carrying, the two pieces can be secured sideways to one another. The other chief differences between the two models are in the sight, hand grip, stock, firing mechanism, and flash deflector. The launcher M9 has no face guard. (See fig. 49① and ②.)

b. Component parts. (1) SIGHTS. The launcher M9 has no front sight. The optical ring sight provides a simple means for determining elevation settings and for estimating leads in tracking moving targets. The sight, which is screwed into the sight mounting bracket, consists of a lens frame with a lens and lens cover, range indicator, and range scale. The hinge stud permits the sight to be folded against the sight bracket when not in use or to be snapped into extended position for use in firing. The lens cover can be snapped over the front of the lens to protect it. The sight rotates on its indicator arm stud to provide for elevation. The indicator arm, which clicks into detent on the range scale, holds the sight in position. The lens of the sight gives the impression of having concentric rings with dark transparent lines crossing at right angles in the center of the field. The horizontal line is used to maintain elevation when aiming at a moving target and the concentric rings are used in estimating the lead. The range scale is secured to the sight mounting bracket by two screws. The sight can be zeroed by loosening the screws and moving the range scale up or down as desired. By loosening the nut on the bottom of the sight it is possible to set the sight for deflection. The range scale is graduated in 50-yard inter-

vals with each 100 yards numbered. The numbers run from 0 to 7, corresponding to a range from 0 to 700 yards.

NOTE: Some M9 launchers will be issued with a bar sight instead of the optical ring sight.

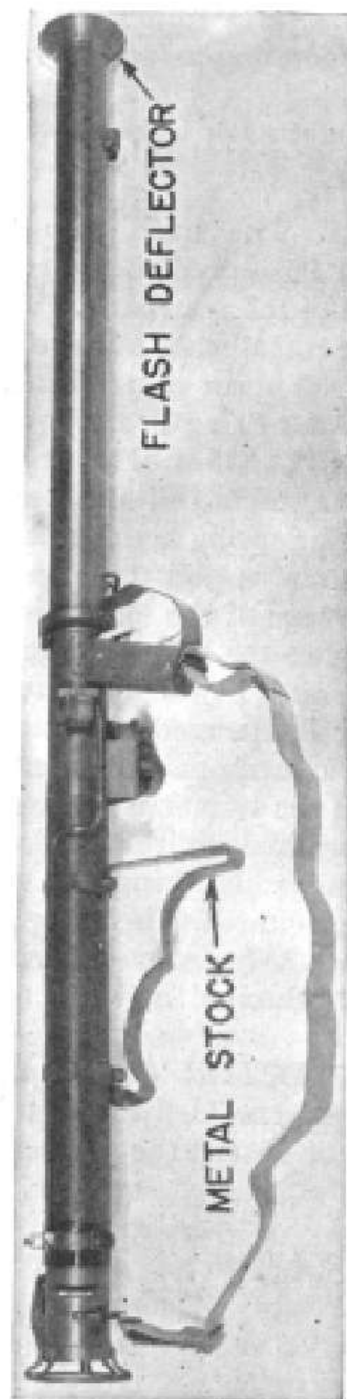
(2) **HAND GRIP.** The hand grip houses the firing mechanism which is secured to the trigger grip support on the under side of the rear barrel.

(3) **STOCK.** The metal stock has two shoulder rests; the front rest is used for short range firing, the rear rest for longer ranges and for firing in the prone position.

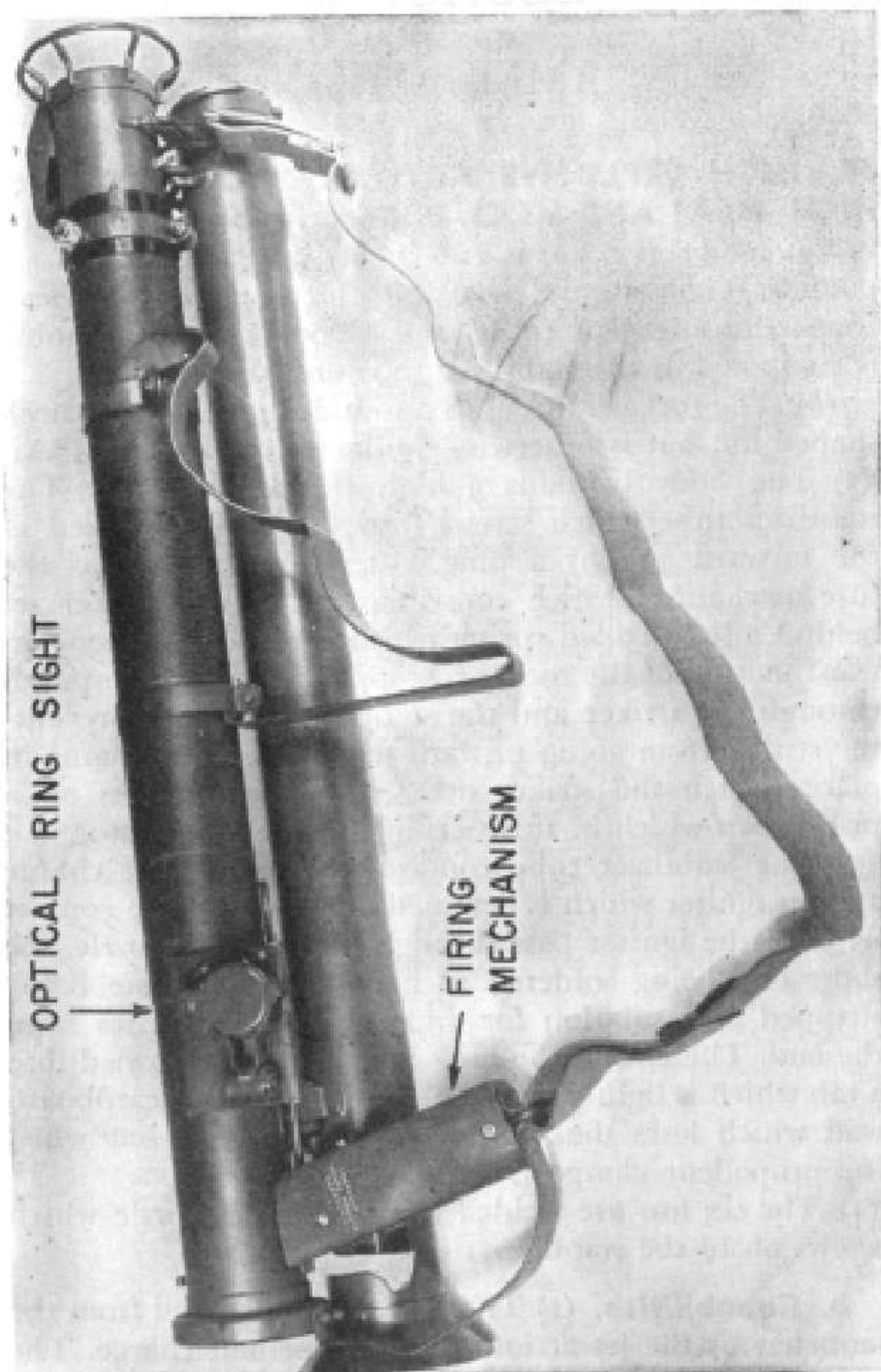
(4) **FIRING MECHANISM.** The firing mechanism consists of magnets housing a coil and a spring actuated bar within the coil. The spring actuated trigger is pinned to the trigger grip support and the firing mechanism. The circuit for the passage of the electric current is complete at all times, but when the trigger is in its normal position, no current is generated by the magneto firing mechanism. When the trigger is squeezed, the bar within the coil is rotated and electric current is generated and flows through the circuit. When the trigger is released, the bar magnet is rotated to its original position under the action of the bar magnet spring and during this movement electric current is again generated. The firing mechanism is provided with a safety. When the safety is put at **SAFE**, the electric current is shunted and the rocket is prevented from firing.

(5) **FLASH DEFLECTOR.** The muzzle flash deflector is of solid metal, spot welded to the tube, with the cone end of the deflector toward the gunner.

Figure 49. Launcher M9.



© Right side.



⑤ Left side.

SECTION III

AMMUNITION

68. HIGH EXPLOSIVE ANTITANK ROCKETS, 2.36-INCH, M6A1 AND M6A3. a. Description.

(1) (a) The rocket M6A1 (fig. 50) is 21.6 inches long and weighs 3.4 pounds. It consists of a head and a fin assembly. The head consists of an ogive (nose) and a body; the fin assembly is composed of the stabilizer tube and the fins.

(b) The rocket M6A3 has a rounded nose and a wheel-shaped fin, but is otherwise similar to the rocket M6A1.

(2) The body contains a high explosive charge. The stabilizer tube which screws into the body is closed at the forward end by a plug. Ahead of the plug is the fuze mechanism which consists of an inertia striker set behind a light coiled spring which is collapsed when forward motion of the rocket is stopped. A safety pin passes through the striker and the stabilizer tube and prevents the striker from going forward while the pin remains in place. When the striker does go forward it sets off a primer cap which in turn sets off the base detonator.

(3) The stabilizer tube contains the propellant charge and an igniter which is actuated electrically. The contact wires of the igniter pass through the Venturi nozzle, the short one being soldered to a fin, the long one being stripped of insulation for approximately 6 inches from the end. The end of the long contact wire is formed into a tab which is lightly taped to another fin. The cardboard wad which loses the Venturi nozzle is blown out when the propellant charge is ignited.

(4) The six fins are welded to the Venturi nozzle which screws on to the stabilizer tube.

b. Capabilities. (1) The rocket is discharged from the launcher by the jet action of the propellant charge. The

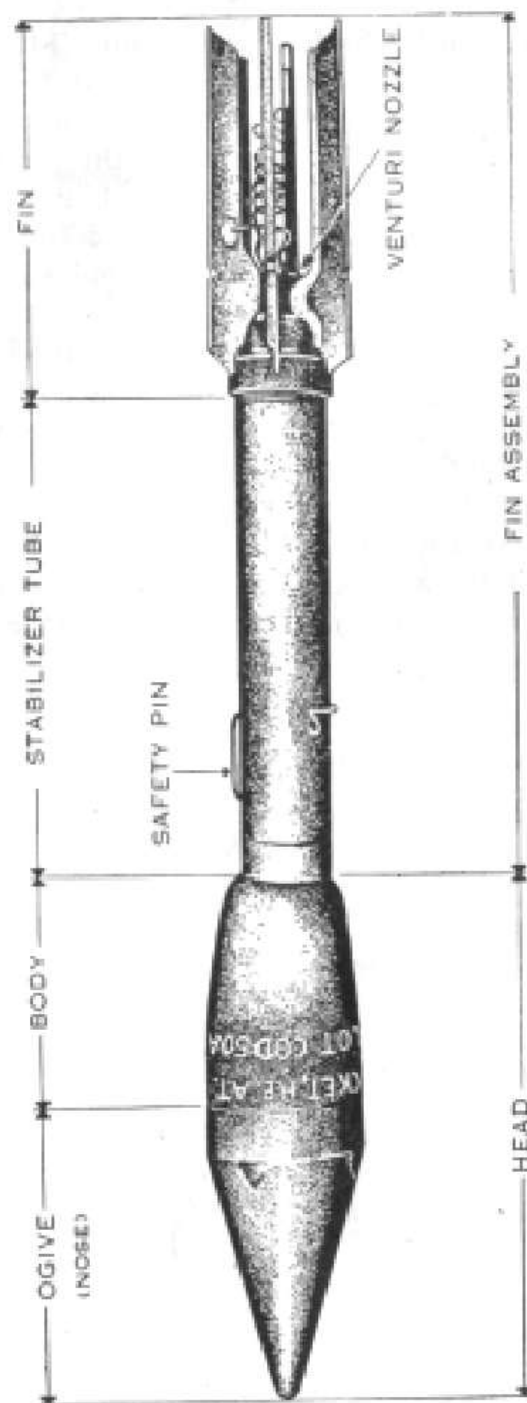
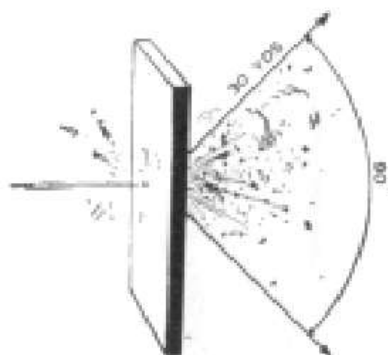


Figure 30. High explosive antitank rocket, 2.36-inch, M6A1.

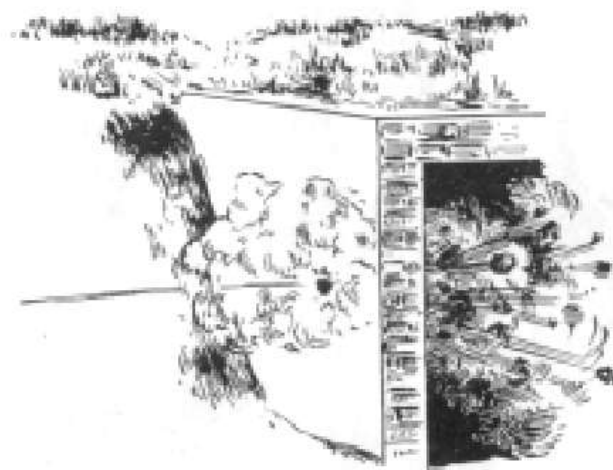
muzzle velocity is only 265 feet per second. The destructive effect of the rocket is produced entirely by the high explosive charge contained in the body, and only sufficient force to operate the fuze is necessary at the point of impact. If the safety pin has been removed, the fuze will operate and detonate if the rocket receives a blow equivalent to dropping it on its nose on normal soil from a height of 48 inches.

(2) The rocket penetrates comparatively thick armor plate at all ranges and at angles of impact as great as 60° from normal or head-on; it blows a hole about 1 inch in diameter through the plate. Particles of the armor, heated to incandescence, are blown from the inside of the plate in a cone of roughly 90° ; any ammunition within the cone is usually exploded. The fragmentation of the body of the rocket occurs outside the plate. (See fig. 51.)

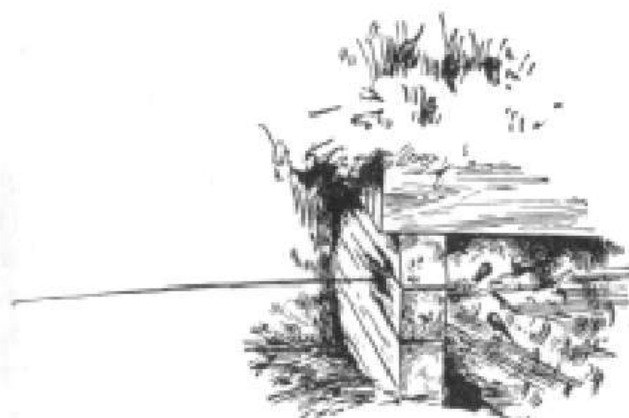
Figure 51. Effect of rocket.



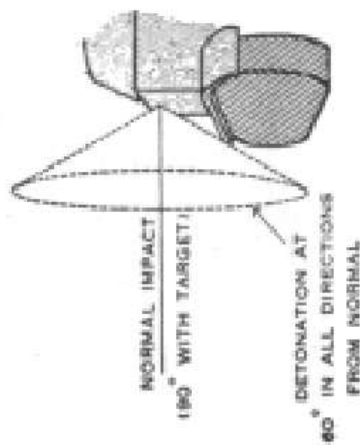
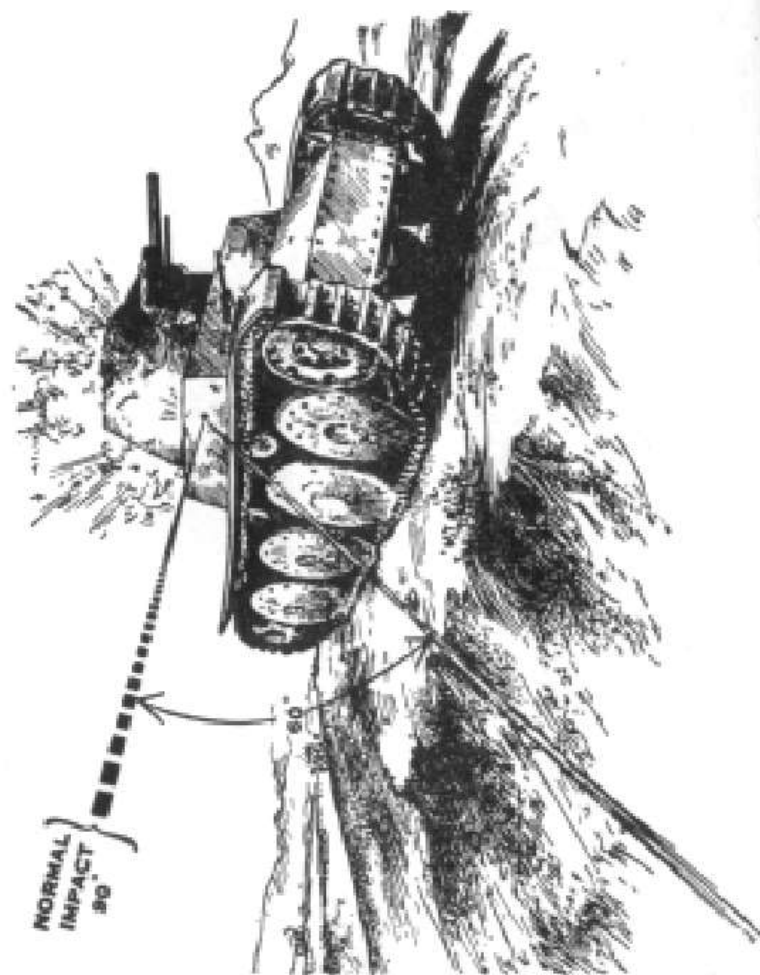
① *Armor plate.*



① *Masonry.*



② *Timbered construction.*



③ Penetration at angles at 60° from normal 100%

(3) It will penetrate brick or rock masonry, or timbered construction up to a considerable thickness.

c. Limitations (fig. 52). The weapon has a relatively low rate of fire; it has a distinctive flash on discharge, tending to disclose the position; and when fired at point targets beyond 300 yards range, it is comparatively inaccurate. Its accuracy when fired at moving targets decreases as the range increases. When fired on flat terrain at less than 300 yards range, impact with the ground ordinarily causes a ricochet and failure to detonate. At any range, impact with mud, soft sand, or water will not produce detonation.

69. PRACTICE ANTITANK ROCKETS, 2.36-INCH, M7A1 AND M7A3.

a. The rocket M7A1 (fig. 53) is identical in size, weight, and external appearance to the rocket M6A1 but contains no high-explosive charge in the body. It has the same propellant charge in the stabilizer tube for discharging it from the launcher.

b. The rocket M7A3 is similar to the rocket M7A1 but has a rounded nose and a wheel-shaped fin.

70. IDENTIFICATION AND MARKINGS. a. Method.

Ammunition is identified by the painting and marking on the item and all its packings. Complete identification of an item of ammunition consists of —

(1) The standard nomenclature which includes type, size, and model designation.

(2) The ammunition lot number which, in general, includes the loader's lot number, the loader's symbol, and the date of loading.

b. M6A1 and M6A3 HE rockets. These rockets are painted olive drab with type, model, and lot number marked in black.

c. M7A1 and M7A3 practice rockets. These rockets are painted black with type, model, and lot number marked in white.

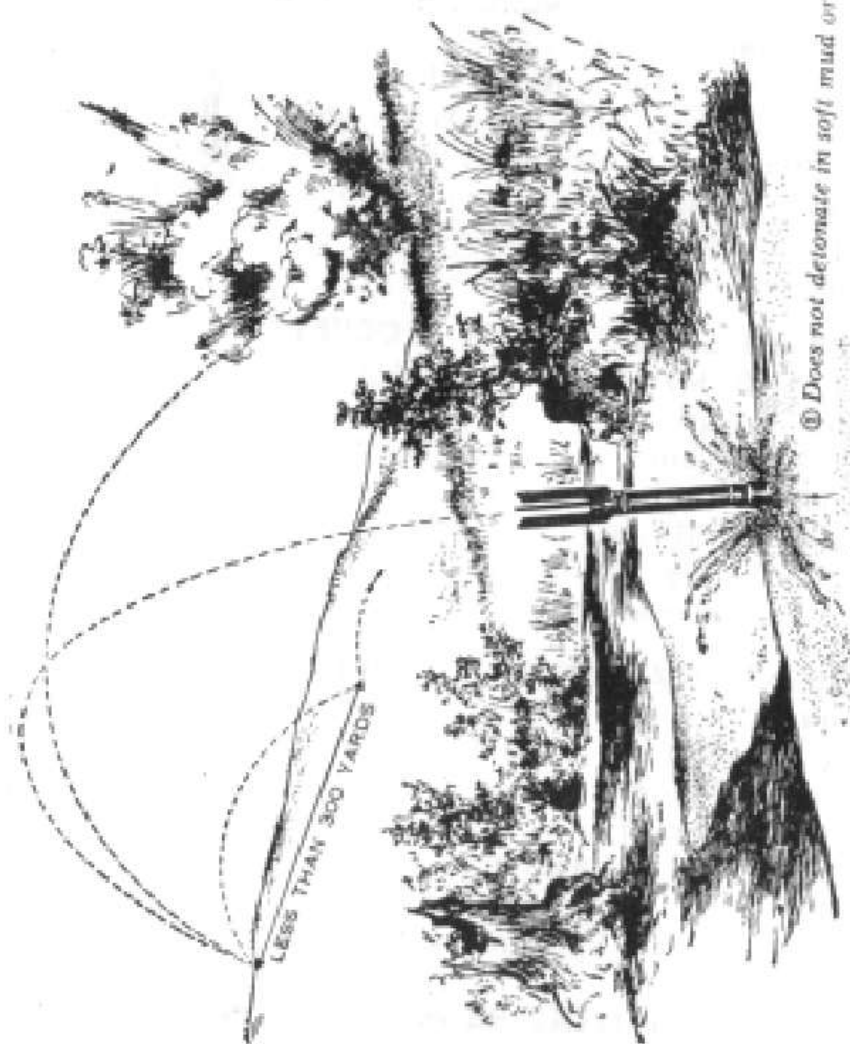


Figure 52. Limitations of rocket.

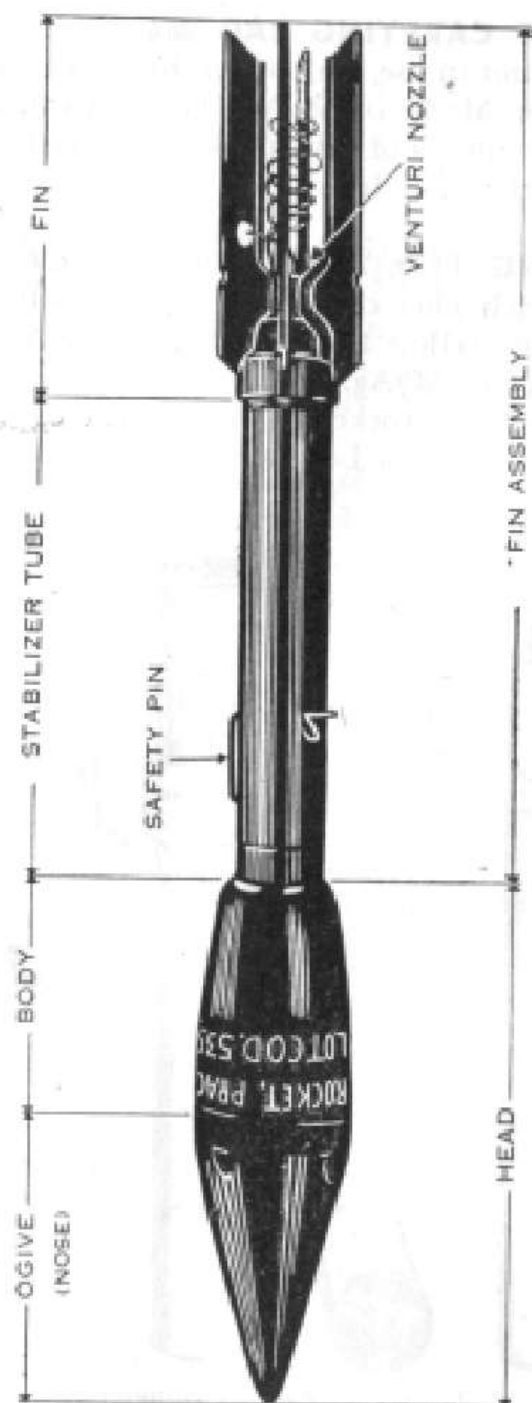


Figure 53. Practice antitank rocket, 2.36-inch, M7A1.

71. ROCKET CARRYING BAG M6. This bag, easily stored when not in use, is designed to carry three rockets, M6A1, M6A3, M7A1 or M7A3. The bag can be carried at the side by means of side handles or over the shoulder by means of the shoulder strap.

72. PACKING (fig. 54). Rockets are packed one per fiber container. Each fiber container is sealed with a strip of adhesive tape (yellow for the M6A1 or M6A3, and blue for the M7A1 or M7A3, practice) printed with the type and model of the rocket. Twenty such containers are packed to the wooden box, weighing approximately 128 pounds.

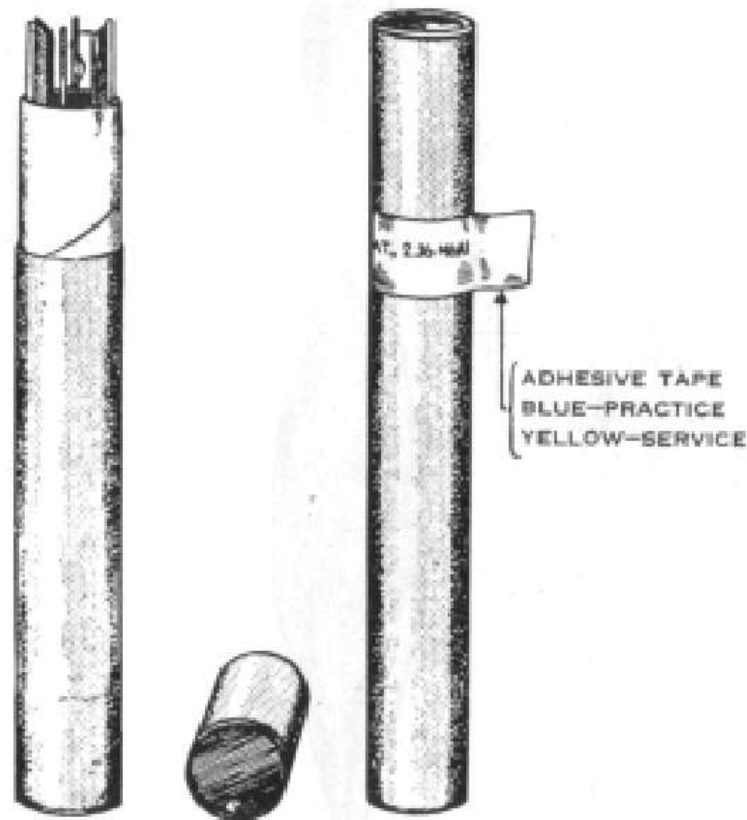


Figure 54. Packing container.

SECTION IV

MAINTENANCE

73. DISASSEMBLY AND ASSEMBLY. a. General. For ordinary care and cleaning, it is not necessary to disassemble the launcher except to remove the trigger grips. Further disassembly is for the purpose of inspection, maintenance, and repair and should be performed under the supervision of an *armorer-artificer or ordnance personnel*.

b. Disassembly launcher M1A1 (fig. 55, 56, and 57).

- (1) Unscrew the screw from the flash deflector mounting clamp and remove the deflector.
- (2) Disconnect the sling from the sling loops on the launcher.
- (3) Unscrew the screws and remove the sling bracket from the trigger grips. Unscrew the screws on the left trigger grip and remove the trigger grips. Remove the cotter pin, withdraw the trigger pin, and remove the trigger. Break the soldered connection between the wire and the lower switch contact support. Pry the conduit with the wire loose from the clips on the left side of the stock support.
- (4) Unscrew the screws from the circuit indicator cover. Remove the cover and break the soldered wire connection. Remove the spare bulb and the socket assembly with bulb.
- (5) Open the hasp on the under side of the stock and remove the batteries. Remove the four screws from the hasp and remove the hasp. Break the soldered connection between the wire and the bottom of the stock support. Slide the stock off the support.
- (6) Press down on tail latch body remove cotter pin and tail latch pivot pin, and carefully withdraw the tail latch spring.

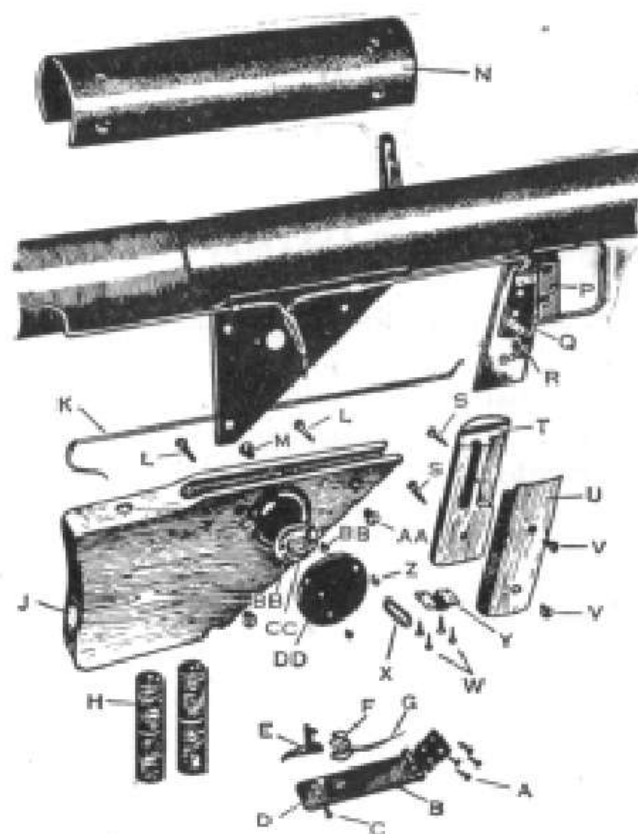


Figure 55. Stock and grip group, launcher M1A1.

c. Assembly, launcher, M1A1. (1) Place tail latch spring in position over the stud between tail latch shunt and tail latch body. Press down on tail latch body and insert the tail latch pivot pin and secure it with the cotter pin.

(2) Solder the broken connections between the wire and lower switch contact support, between the wire and upper stock contact, and between the wire (on the battery spring) and the bottom of the stock support.

(3) Slide the stock partly onto the stock support. Keep the wire between stock support and battery spring moving in the slot in the stock so as not to break the soldered connections. Replace the conduit with the wire in the

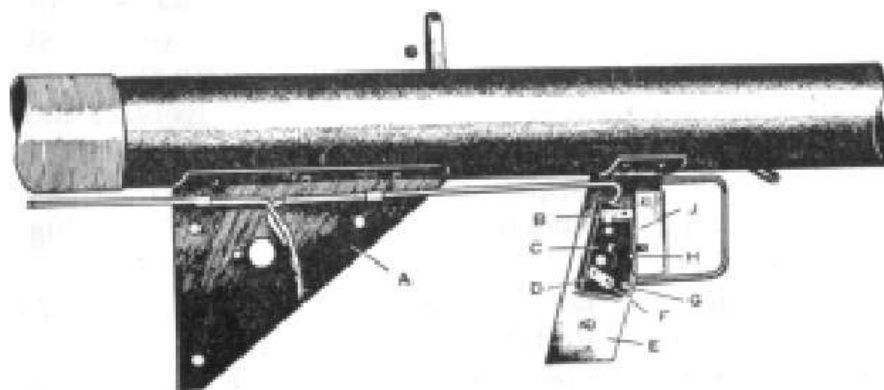


Figure 56. Trigger group, launcher M1A1.

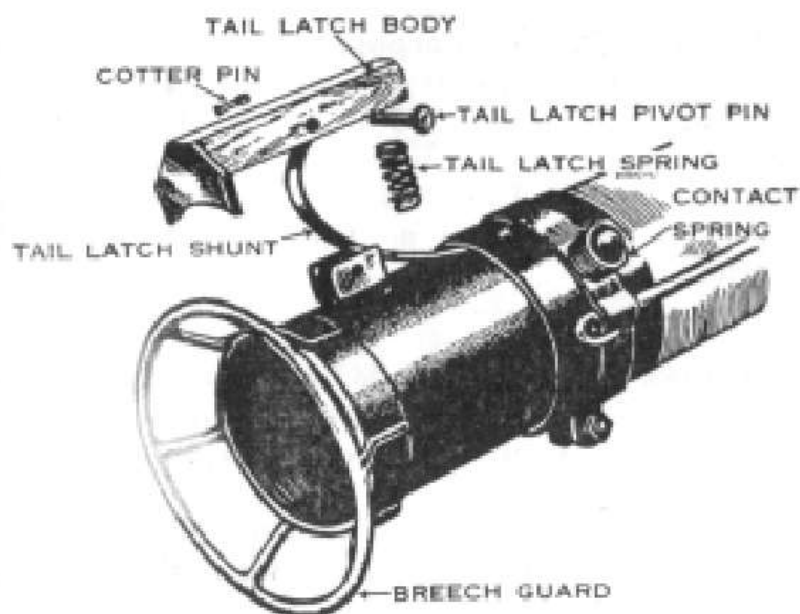


Figure 57. Rear of launcher M1A1 or M9.

clips on the left side of the stock support. Push stock into position on the stock support.

(4) Assemble light bulb to miniature socket. Place unit in circular hole in stock and secure socket with two screws. Pull the socket wire out through the hole in top of stock and solder to the wires leading from the conduits on right side of stock support. Replace circuit indicator cover and secure with the three screws. Attach the battery spring to the hasp and assemble the hasp to the stock. Replace the batteries in their compartments.

(5) Attach the trigger grips to the trigger support. Attach the sling bracket to the trigger grips and secure the sling to the launcher.

(6) Slide the flash deflector over the muzzle of the barrel and secure in place by the mounting clamp.

d. Disassembly and assembly, launcher M9. Instructions on disassembly and assembly of launcher M9 will be published later.

74. INSPECTION. a. Purpose. Thorough, systematic inspections and prompt repairs are insurance that the weapon will not fail at a critical moment. Inspection discloses the repairs and adjustments necessary to insure proper functioning of the weapon. An inspection should be made of the launcher at delivery and periodically thereafter.

b. Procedure, launcher M1A1. (1) Inspect the launcher for its general condition, loose or broken components, chipped paint, bends, dents, or obstructions in the bore.

(2) Test the electric circuit by squeezing the trigger several times. The bulb should light when the trigger is squeezed.

(3) Check for any loose wire connections.

(4) Remove the trigger grips and check for the following:

- (a) Deformed or broken switch contact bar.
- (b) Missing switch contact button.
- (c) Missing bar contact.
- (d) Missing switch contact lower support rivet.
- (e) Loose connections, particularly between switch contact bar and upper switch contact support; and between wires and the upper and the lower switch contact supports.

(f) Corroded electric contact points.

(5) Remove the circuit indicator cover and check for loose wire connection. Remove the hasp and check for loose connection between wire and stock support.

(6) Remove the batteries and examine for condition. Replace them if cracked, swollen or broken.

(7) Check functioning of tail latch and of trigger spring.

(8) Examine the barrel of the launcher for rust, scale, and dents.

(9) Examine the contact springs to see that they are secure to the mounting clamp and that they are clean and free from rust, paint, and grease.

c. Procedure, launcher M9. Instructions on inspection procedure for launcher M9 will be published later.

75. PREVENTIVE MAINTENANCE. The following operations will be performed by the using arms:

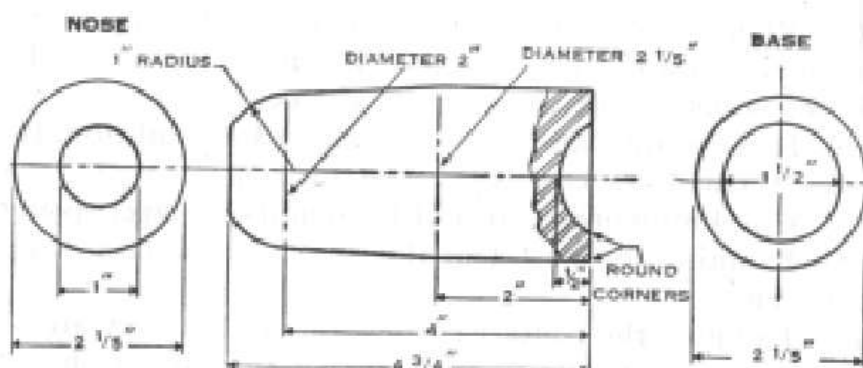
a. Tighten all screws and nuts to prevent their becoming loose in service. This should be done periodically when the weapon is in service. Clean all rust and pitted surfaces with crocus cloth and repaint with lusterless olive-drab enamel.

b. If the switch contact bar is deformed, it may be bent to its proper shape, but it should be replaced at first opportunity.

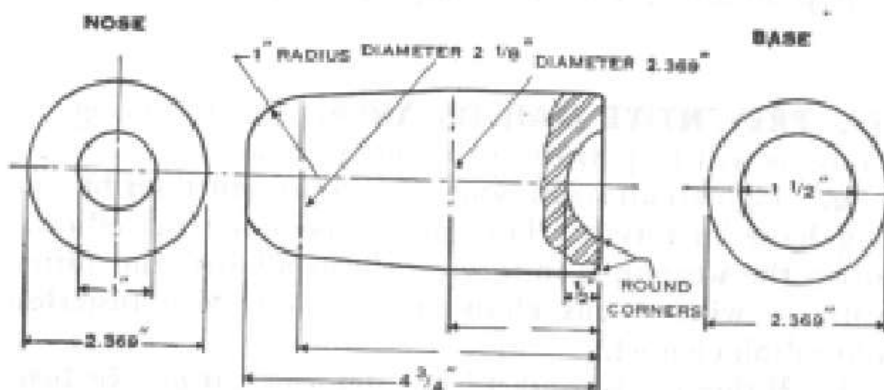
76. REPAIRS (fig. 58① and ②). The company armor-artificer or ordnance maintenance personnel will make the following repairs:

- a. Solder all loose or broken soldered connections.
- b. Replace any defective bulbs.
- c. Replace the trigger assembly if the trigger spring is too weak or if trigger stop is missing.

Figure 58. Hardwood punch for removing dents.



① Small dent remover, 2.187 inch diameter.



②

② Large dent remover, 2.369 inch diameter.

d. If tail latch does not function, replace the tail latch spring.

e. If trigger grips are broken or cracked allowing moisture to enter, replace grips.

f. If electrical contact points are corroded, clean with crocus cloth.

g. Remove dents in the barrel by using the tools shown in figure 58. These tools should be improvised.

(1) Drive the small dent remover through the barrel. This will increase the diameter of the dented part of the barrel up to 2.187 inches.

(2) Drive the large dent remover through the barrel. This will bring the diameter of the dented part of the barrel back to the original 2.36 inches.

(3) To restore the original round contour of the barrel, force the dent remover to place under the dents. Using the dent remover as an anvil, hammer the outside surface of the barrel with a ball peen hammer.

h. Additional instructions on repair of launcher M9 will be published later.

77. CARE AND CLEANING OF LAUNCHER. a. Daily and immediately after firing.

Wet a cloth with rifle bore cleaner and run it through the barrel several times. Then run a dry wiping cloth through until the inside of the tube is clean and dry. Saturate a clean dry wiping cloth with light preservative lubricating oil, wring it out and oil the bore by running the cloth back and forth through the barrel.

b. Weekly. Apply one or two drops of light preservative lubricating oil to the tail latch pivot pin.

c. Prior to firing. Run a clean, dry wiping cloth through the barrel until, by the appearance of the cloth, it is indicated that the film of oil has been removed from the bore.

d. External parts. Rusted, pitted, or chipped exposed metal surfaces should be cleaned with crocus cloth and repainted with lusterless, olive-drab, synthetic enamel. Do not paint the contact springs, flash deflector on launcher M1A1, or that part of tail-latch which makes electrical circuit with rocket complete. To clean wooden stock and grips, wipe with a clean wiping cloth lightly oiled with Type A raw linseed oil. Wipe off thoroughly with a *clean, dry* wiping cloth.

e. When used in wet weather. Disassemble the stock and trigger grips and dry all electric contact points and wires with a dry cloth. Clean and oil the rest of the launcher as described above.

78. UNUSUAL CLIMATIC CONDITIONS. When operating under unusual conditions such as tropical or arctic climates, severe dust or sand conditions, and near salt water, it is essential that all the precautions listed below should be observed.

a. Arctic climates. In temperatures below freezing, but particularly in arctic climates, all operating parts should be kept absolutely free of moisture. The bore of the launcher should be cleaned daily with a clean, dry cloth and oiled with special preservative lubricating oil. The batteries should be removed from the launcher M1A1 and kept warm until just before firing. Carrying batteries in inner pockets will keep them sufficiently warm. Immediately upon bringing indoors, the launcher should be cleaned on the outside and inside with a dry, clean cloth. Remove the grips, clean and dry the launcher again, and oil the bore.

b. Tropical climates. (1) When temperature and humidity are high, when salt air is present, or during the rainy season, the launcher should be thoroughly inspected and cleaned daily. The bore should be oiled with medium preservative lubricating oil somewhat more lib-

daily than prescribed in paragraph 77a. Wood parts that have become swollen due to moisture should be shaved off sufficiently to relieve binding. A light coat of raw linseed oil applied at least once a month and well rubbed in with the heel of the hand will help to keep moisture out. Allow oil to soak in for a few hours and then wipe and polish the wood with a dry, clean rag. Care should be taken to see that linseed oil does not get onto electrical contacts as it will gum when dry.

(e) In hot, dry climates where sand and dust are likely to get into the bore, the launcher should be wiped clean daily, inside and out. Oiling of the bore should be done sparingly. In such climates, wood parts are apt to dry out and shrink, and more frequent application of raw linseed oil will help keep wood in condition. During sand or dust storms, the breech and muzzle should be kept covered.

79. PRECAUTIONS. a. Steel helmets should be worn by the gunner and loader.

b. For protection from the flash, the gunner should wear gloves and goggles or similar protective equipment such as a cloth wrapped around the hands and a gas mask to shield the face.

c. After the safety pin has been removed, the rocket should be handled with more than ordinary care due to the sensitivity of the fuze.

d. When the propellant is ignited, gases and flames are blown from the breech of the launcher. Therefore, to avoid injury when firing from the prone position, the gunner must place his body so that it is at an angle of 45° to the line of aim.

e. The area directly in rear of the launcher for a distance of 20 feet must be clear of personnel or of inflammable material.

f. The rockets should not be fired at temperatures below 0° F. nor above 120° F.

g. At temperatures below 14° , the batteries in the launcher M1A1 are so weakened that they will not ignite the propellant charge. When operating in such temperatures, batteries should be removed from the launcher and kept warm until just before firing by carrying them in inside pockets. The spares may be carried in the pockets and switched with the cold ones every hour.

h. The rocket as shipped contains a cardboard wad cemented over the Venturi nozzle which prevents entry of moisture or dirt into the propellant charge. Do not use any rocket with a missing cardboard wad. Do not remove the wad.

i. During the operations of loading and sighting, the gunner should always keep the muzzle of the launcher pointed toward the target.

j. If the rocket fails to fire and if the corrective measures taken (par. 83b) indicate that the fault does not lie in the electrical circuit of the launcher, on command, remove contact wire from contact spring, raise tail latch, and carefully withdraw rocket from the launcher until the safety pin can be reinserted. Reinsert safety pin, remove rocket completely from the launcher and place it to one side for disposal by authorized personnel.

k. Always set safety of launcher M9 at SAFE before unloading the launcher.

l. No attempt should be made to fire from the launcher M1A1 or M9 the earlier rocket model M6 and M7 identified by the contact ring on the nose (ogive) and a connective wire taped to the outside running from this ring to the tail.

m. Boxes containing rockets should not be dropped, thrown, tumbled, or dragged.

n. The waterproof seal of the fiber containers should not be broken until the ammunition is about to be used. Items unpacked but not used should be restored to their original condition and packing, and resealed.

o. Rockets should be protected against moisture and extremes of temperature. They should not be stored where temperatures exceed 120° F.

p. Place rockets in their containers in shaded places. Do not expose rockets to direct rays of the sun until they are ready to be loaded into the launcher.

q. Do not allow a rocket to remain in a hot launcher prior to firing.

SECTION V

OPERATION

80. GENERAL. **a.** Due to the comparatively short sight radius, care should be taken to assume correct firing positions (see par. 86*d*).

b. The weapon, especially when the tube becomes heated from firing, has a tendency to shoot high, making it necessary to aim at the bottom of the target.

c. Interpolation between the ranges provided on the front sight, estimation of ranges in excess of 300 yards and of windage adversely affect accuracy of fire.

81. LOADING. **a.** *During the operations of loading, sighting, and firing, the loader should at all times be to the side of and slightly forward of the breech end of the launcher.*

b. The sequence of loading is as follows (fig. 59):

(1) The gunner places launcher on right shoulder and tests the electric circuit by squeezing trigger several times, observing to see that test lamp lights. (There is no test lamp on the launcher M9.)

(2) The gunner takes the prone, sitting, kneeling, or standing position with the launcher on his shoulder. The loader takes a similar position facing the launcher, op-

posite the gunner, and as close to the launcher as possible. The loader should be approximately 1 foot from the launcher and little, if any, to the rear of the gunner.

(3) The gunner commands: LOAD, at the same time dropping his right hand from the trigger and tapping the loader, while supporting the launcher with his left hand. The gunner makes certain the test lamp (on launcher M1A1) is not lighted. He must not squeeze the trigger while launcher is being loaded.

(4) The loader repeats the command, shifting his weight to the left without changing his position on the ground.

(5) The loader grasps a rocket with his left hand, palm up. The rocket is grasped by the tube, with the head of the rocket to the loader's right. The eye of the safety pin is away from the loader and the middle finger of the left hand rests on the eye. (See fig. 59①.)

(6) Simultaneously, the loader grasps the launcher tube and tail latch with his right hand, palm down. His thumb is around the bottom of the tube and his fingers rest on the front portion of the tail latch. Pressing with the fingers, he raises the tail latch. (See fig. 59②.)

(7) The loader inserts the rocket so that the large part of the head is just past the tail latch. He releases the tail latch and with the middle finger of the left hand flips the eye of the pin downward. If the pin does not fall out of the stabilizer tube, he puts his left thumb under the bent portion of the pin and lifts it out.

(8) The loader then slides his left hand back around the fins while he again raises the tail latch with his right hand. He pushes the rocket into the launcher until the notches in the fins line up with the tail latch which is allowed to drop into a notch or notches. To insure a good electrical contact, he gives the rocket a slight rotary twist with the left hand. (See fig. 59③.)

Figure 39. Sequence of loading.



① Loader grasps rocket in left hand.



② Inserts body of rocket partially into launcher; removes safety pin.



② *Raises tail latch; pushes rocket completely into launcher.*



③ *Removes contact wire from rocket fin.*



③ *Affixes contact wire to contact spring of launcher.*



④ *Taps rocketeer; calls "Up."*

(9) The loader pulls the contact wire from the fin and pulling it straight back to uncoil it, fastens it by twisting it around the contact spring. He then taps the gunner and calls "UP." (See fig. 59④, ⑤, and ⑧.)

82. TO FIRE LAUNCHER. The launcher is fired by squeezing the trigger with a steady pressure (see par. 86c). There is no slack to be taken up as on a rifle, the actual firing being accomplished by electrical contact. When firing the launcher Mg, squeeze the trigger and release it immediately. Do not move the launcher off the target before releasing the trigger.

83. MALFUNCTIONS AND IMMEDIATE ACTIONS. Malfunctions on the launcher may be classified as a failure to load or a failure to fire. Below are some of the more common failures and corrective measures applicable to reduce them in the field.

a. Failure to load.

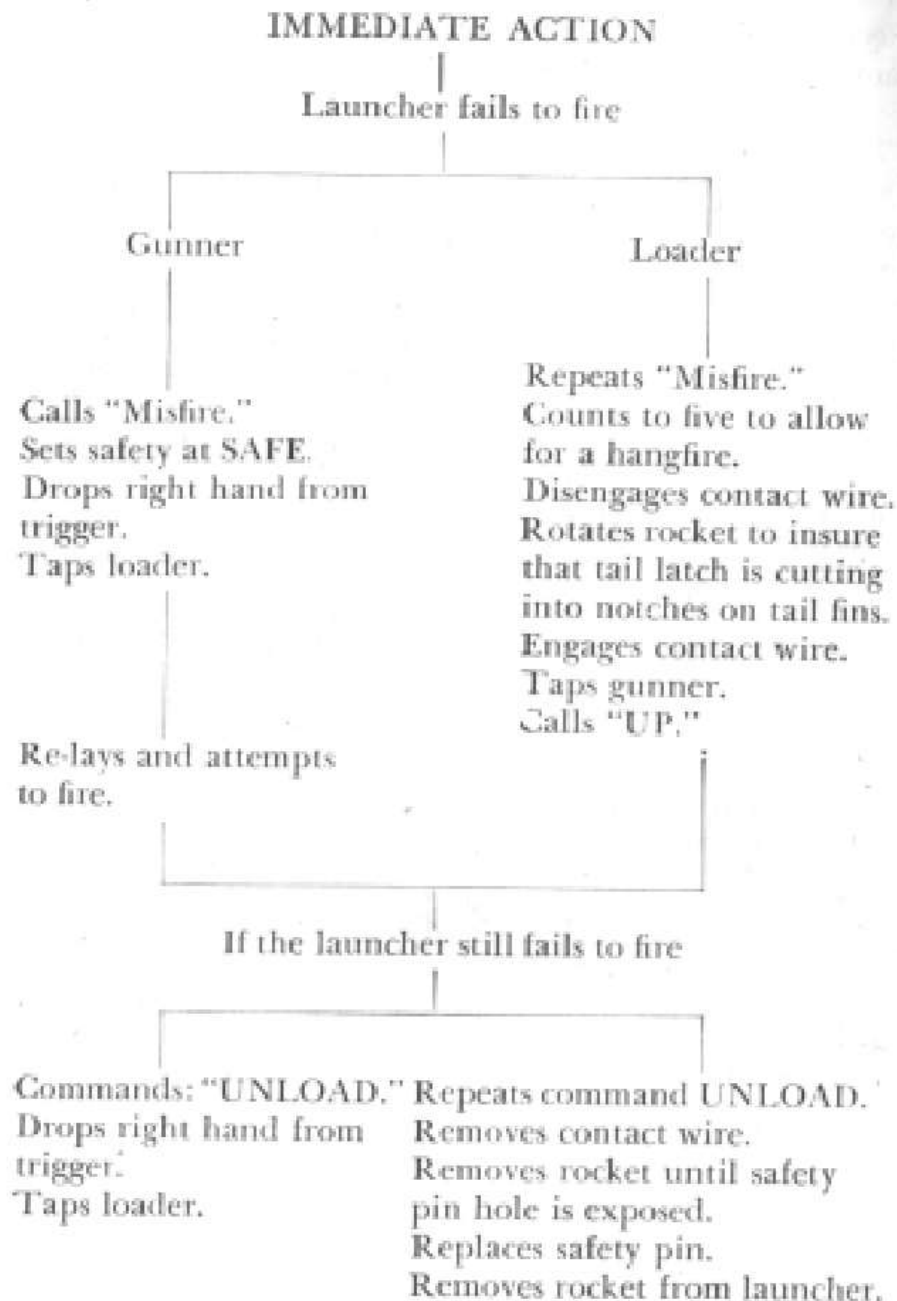
Cause	Corrective measure
Bent tube. Bent breech guard. Oversize or bent round. Bent or defective tail latch.	None. Launcher must be replaced. Straighten with pliers. Discard and get new round. Attempt to straighten latch. If contact cannot be made, the latch will have to be replaced. <i>Caution:</i> Never wedge up forward position of tail latch to get contact.

b. Failure to fire. A failure to fire may be the result of weak batteries (in launcher M1A1) or a break in the electrical circuit. The following are some of the more common causes and the remedies:

Cause	Corrective measure
Weak current. Indicator lamp fails to light, launcher fails to fire.	Change batteries. Change batteries. If launcher still fails to fire, inspect and clean trigger mechanism.
Extreme cold (14° to 0°).	Keep batteries warm until just prior to their contemplated use.
Dampness.	Protect launcher from dampness. If wet, dry thoroughly with clean dry cloth; disassemble as necessary.
Contact wire not securely connected to contact spring.	Attach securely.
Poor contact between tail latch and notches in fins.	Rotate rocket back and forth in order to seat the tail latch firmly on the notch.
Loose or broken parts in trigger mechanism.	Remove trigger grips and check trigger switch contact. If contact bar is deformed bend it into correct shape; if broken, turn launcher over to ordnance maintenance personnel.
Coiled battery spring on hasp fails to contact batteries.	Spread spring coils so that it will make contact.
Loose wires in circuit indicator recess.	Remove circuit indicator cover and tighten wires.
Loose contact wire between battery spring and stock support.	Turn launcher over to ordnance maintenance personnel.

NOTE: The above references to batteries apply only to launcher M1A1.

c. Immediate action. The procedure for immediate action is shown in the following diagram:



SECTION VII

TACTICAL EMPLOYMENT

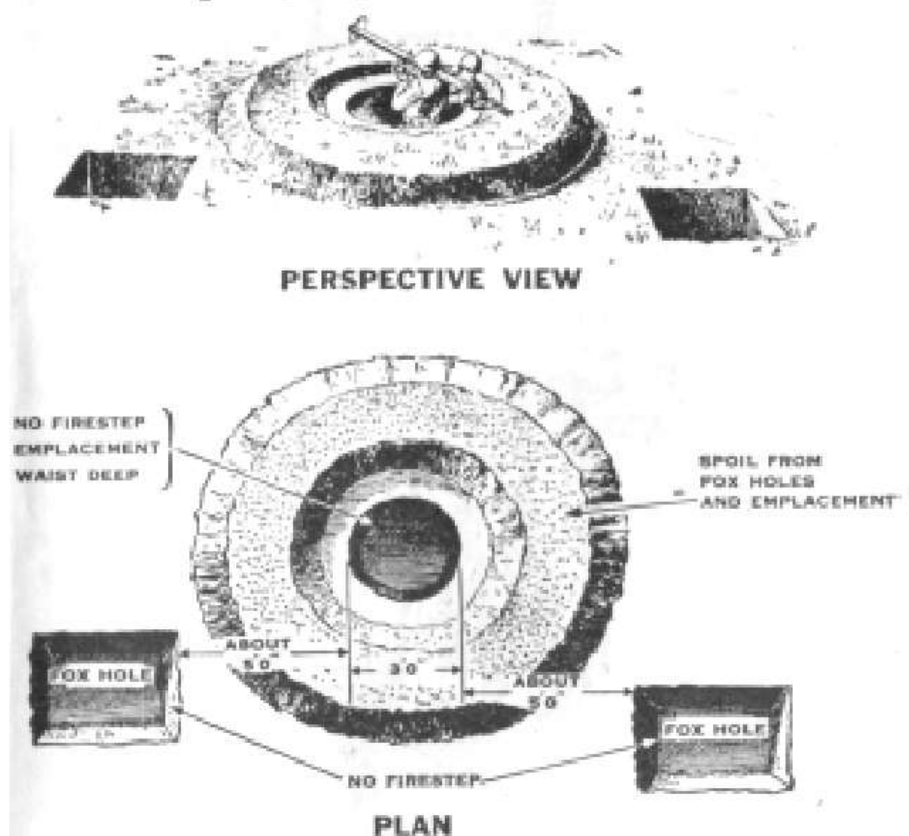
89. GENERAL. The tactical employment of the rocket is governed by its capabilities and limitations. It has its principal function as a defensive antimechanized weapon, although tank destroyer personnel, motorized reconnaissance units, and other specially organized groups may use it offensively in many situations. The rocket does not supplant but supplements the antitank rifle grenade M9A1 and such other antimechanized weapons as may be issued to any unit.

90. OFFENSIVE ACTION. The rocket is an effective weapon for use by raiding groups, tank hunting parties, reconnaissance elements, and in the attack of towns and villages. The rocket is especially effective in attacks on vehicular bivouacs, halted or disabled vehicles, and for use in ambushes. It is capable of delivering harassing fire against an area target up to ranges as great as 650 yards. However, ammunition must be conserved in order that the rocket may perform its primary mission — defensive action against mechanized forces. Being highly mobile, the anti-tank rocket is a valuable weapon with any landing force which may expect to encounter pillbox or mechanized defenses. In such operations, specially organized teams should accompany the leading elements of the landing force.

91. DEFENSIVE ACTION. For the close-in defense of units and installations, the rocket is a highly mobile and effective weapon. Normally, rocket gunners and loaders

will be designated from among ammunition bearers, truck drivers, clerks, and other personnel. However, when required by the situation, all launchers must be manned by any available personnel who have been trained in their use. Whenever time permits, rocket teams will be assigned a definite place in the antimechanized defense of the weapon, unit, or installation. If natural cover against mechanized attack is not available, pit type or pit fox hole type emplacements should be dug for the rocket teams. (See fig. 65.) In case of surprise attack by hostile mechanized forces, gunners will direct their fire against the nearest

Figure 65. Emplacements for rocket team.

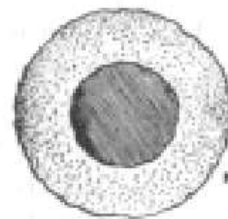


② Pit fox hole type.

hostile vehicle. In the absence of orders or standing operating procedure, gunners will maneuver so as to place themselves in the most favorable position to engage and destroy hostile mechanized vehicles in the immediate vicinity which constitute a threat to the unit or installation. The normal tactical employment of the rocket as a defensive weapon is enumerated below:

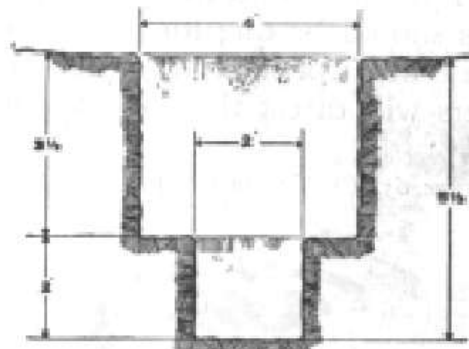
- a. Close-in defense of crew-served weapons.
- b. Protection of motorized columns on the march and at temporary halts. Under these conditions, it must be noted that it is not desirable to fire rockets from a vehicle, because of the blast effect of the propellant on occupants of the vehicle, its cargo, and the vehicle itself. In all cases the launcher must be readily available for use by dismounted action (par. 93).
- c. Protection of mine fields and wire entanglements.
- d. Protection of observation and command posts.
- e. Defense of rear area installations of all arms and services within range of hostile mechanized forces. These will include —
 - (1) Train bivouacs.
 - (2) Ammunition distributing points.
 - (3) Supply installations of all types.
 - (4) Landing fields and airdromes.
 - (5) Critical points on a route of hostile mechanized advance which may not be protected by other adequate means.
 - (6) Higher headquarters installations.

92. SPECIAL TACTICAL EMPLOYMENT. Using arms and services should develop standing operating procedure peculiar to their needs for the most appropriate employment of this weapon. To amplify the previously mentioned general tactical considerations and the normal employment of the rocket, the following procedure will govern when applicable:



NOTE:
SPOIL
REMOVED

PLAN



SECTION



© Pit type.

2. Personnel. Whether or not all men have had an opportunity to fire the rocket launcher, they should have at least received instruction in the operation, use, and maintenance of the weapon, together with the real thing concerning its capabilities and limitations. Subsequent trainers will insure that all available weapons are promptly replaced as one dies.

b. Defense of close-armed weapons. (1) The use and use of converted weapons are particularly subject to mechanical attack. Close-armed weapons which have difficulty in making rapidly moving targets or targets which pass rapidly in the line of vision automatically ineffective against armored vehicles, therefore, provided in the design and price in rifle grenades and rocket grenades. And therefore, these weapons are not always considered necessary for the service of the plane should be employed in grenades and rocket grenades when under close of mechanical attack. I shall discuss the appearance of a new vehicle should be — mechanical, and position as far as possible in grenades — and rocket grenades should and provided as well as these weapons should be made.

[30] Many languages (e.g. English) possess a system of secondary clitics that are in their primary focus with a cluster of morphemes that are in focus in the primary focus. The morphemes in the primary focus are in focus in the primary focus. The morphemes in the primary focus are in focus in the primary focus.

4. **Release of masticated volume.** Vehicle, carrying biomass and volume should be distributed throughout the volume. All debris should be approximately 1000 g & masticated solids by increasing up to 10% the diameter of vessel should be 10. If an increase in weight & mass solid solids is not possible, vehicle should be driven to any available road that could provide and provide enough the current local vehicle status. Vehicle can be distributed across. Road and growth equipment.

rockets and launchers in vehicles must permit instant availability for dismounted action. Rockets may be carried in vehicles in the original packing case, or containers of not more than three rounds may be taped or strapped together. The M6 ammunition bag is suitable for carrying rockets when dismounted, the maximum load being three rounds per bag. Ammunition must be protected from dirt and moisture and secured against excessive shocks which might cause detonation.

94. SECONDARY EMPLOYMENT (fig. 66). In addition to its use as a projectile when fired from the launcher, the rocket may be prepared for firing electrically and used as an improvised antitank mine as follows:

a. Remove the rocket from its container. With the top (open) end of the empty container resting on the ground or other surface, punch a hole in the bottom. Insert the ends of the actuating wires in the hole. Twisted-pair field telephone wire is suitable for use as actuating wires. Lift the container with one hand and push the wires through until they extend about 8 inches beyond the open end of the container. This extension allows sufficient leeway to connect the actuating wires to the rocket. A coiled contact wire is lightly taped to one of the rocket fins. Remove this wire from the fin and connect it to one of the actuating wires. Connect the other actuating wire to the wire which is soldered to one of the rocket fins, by wrapping it tightly around the soldered connection. Slide the rocket, fins first, partially into the container, remove the safety pin, and then slide the rocket down all the way. Place a piece of the adhesive container seal over the exit of the actuating wires to help keep out moisture. Close the space between the open end of the container and the rocket head with the remainder of the container seal, for the same reason.

b. Place the assembly in a hole about 2 feet deep, with nose up, or horizontally in the side of a bank or cut, prop-

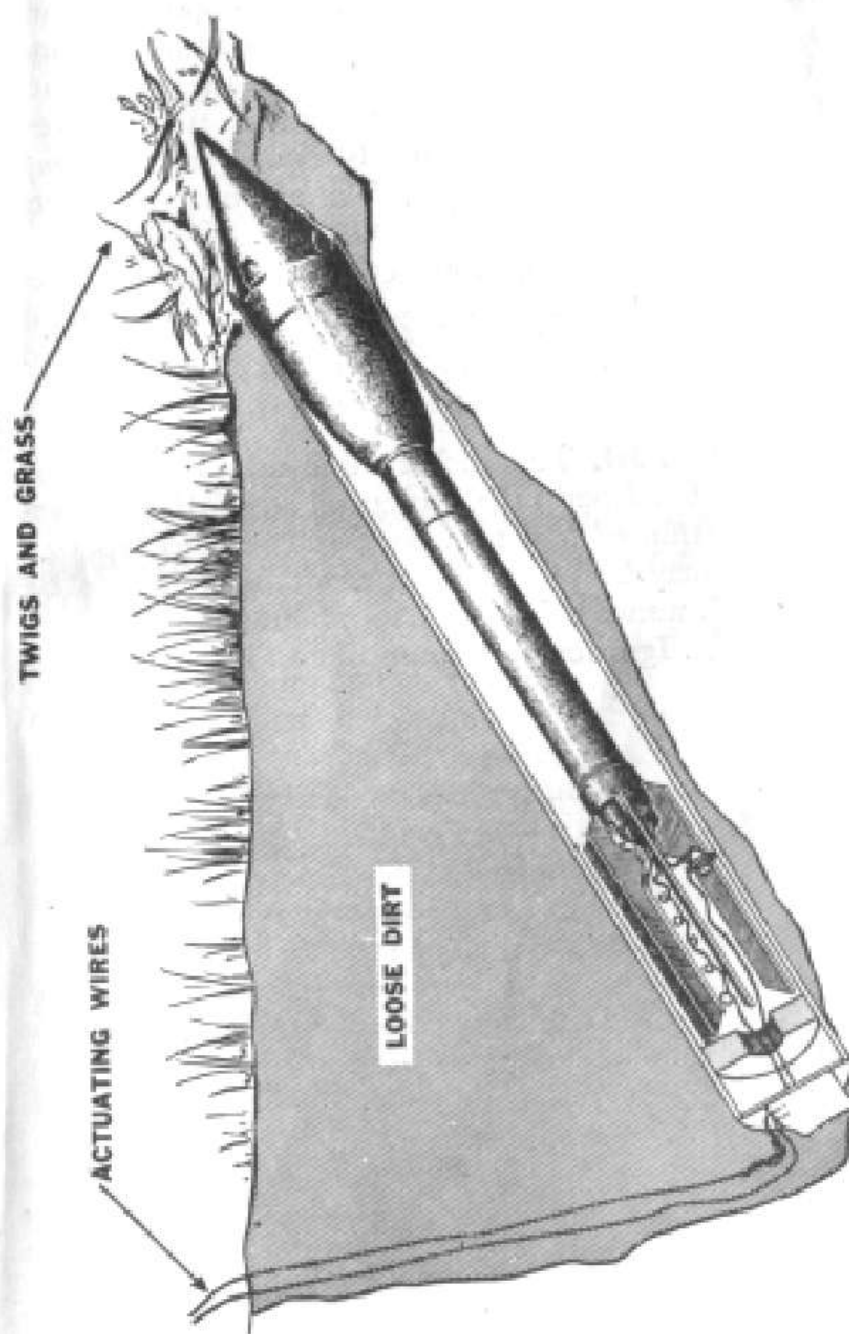


Figure 66. Rocket used as antitank mine.

erly sited to achieve the desired effect. The rocket container serves as a guide for the rocket. If placed in a hole, use loose dirt to cover all of the assembly except the upper end, which should be concealed by covering with twigs, leaves, or light bunches of grass. *Care must be taken not to bump the nose, as the rocket is completely armed once the safety pin is removed.*

c. Arrange a circuit by connecting the actuating wires from the rocket with a battery and an improvised circuit closer. The battery and connections should be protected from moisture.

95. DEMOLITION. The rocket mine may be used for destruction of railroad rails, structural steel, disabled vehicles both armored and transport, and various types of matériel. It may be placed in or near the object to be destroyed, allowance being made for about a foot of travel before impact. Ignition is accomplished as in paragraph 94a.